Green Campus Performance Dashboard

Tech Stack & Architecture

The **Green Campus Performance Dashboard** follows a **MERN-style architecture with a Flask backend**:

- **Frontend:** React + Vite
- **Backend:** Flask (Python REST API)
- **Database:** MongoDB
- **Data Format:** JSON

Justification of Technology Stack:

React.js is used for building a component-based user interface that allows reusable components like sidebar, cards, charts, and tables.

Vite provides a fast development server and optimized build process.

Flask is a lightweight backend framework used to build REST APIs and connect the frontend with MongoDB.

MongoDB stores sustainability data in flexible document format suitable for analytics dashboards.

Recharts is used to visualize energy, water, and waste data through charts.

System Architecture Flow:

User interacts with React Dashboard (Frontend).
Frontend sends HTTP requests to Flask REST API.
Flask retrieves data from MongoDB database.
API returns JSON response to frontend.
Dashboard updates charts, cards, tables, and alerts.

API Endpoints:

/api/energy

/api/water

/api/waste

User (Admin/User)
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▼
React Dashboard (Frontend)
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│ HTTP Request (fetch/axios)
▼
Flask REST API Server
│
│ Query
▼
MongoDB Database
│
│ JSON Response
▼
Flask API
│
▼
React Dashboard Updates Charts & Cards